

Research Proposal 2026

Category: Computer Sciences & Software Development (COM)

Project Type: Engineering / Computer Science

1. Research Question

Can a multi-layered analysis system, integrating **heuristic pattern recognition** and **privacy-preserving breach lookups**, provide a more scientifically accurate evaluation of password strength than traditional rule-based methods?

2. Aim and Objectives

The aim of this project is to design, build, and test a software application (SentinelVault) that evaluates password security through progressively more intelligent stages.

- **Objective 1:** To develop a baseline UI and logic to capture user input securely.
- **Objective 2:** To implement a **Heuristic Analysis Engine** using the zxcvbn library to detect patterns like dictionary words, l33t-speak, and keyboard walks.
- **Objective 3:** To integrate the **Have I Been Pwned (HIBP) API** using the **k-Anonymity protocol** (SHA-1 hashing) to check for leaked credentials without compromising user privacy.
- **Objective 4:** To calculate and display true **Information Entropy (bits)** to provide a mathematical measurement of password strength.

3. Background and Significance

Traditional password checkers rely on simple "Regular Expressions" (e.g., "must have a capital letter"). These are flawed because passwords like P@ssword1! meet the criteria but are extremely easy for computers to crack. By focusing on entropy and historical leak data, this project provides a "Zero-Knowledge" security audit, ensuring users can check their security without ever uploading their plain-text password to a database.

4. Methodology (Engineering Design)

- **Prototype 1 (UI & Layout):** A web-based interface using HTML5 and CSS3 designed with a high-contrast security theme.
- **Prototype 2 (Local Intelligence):** Integration of pattern-recognition logic to move beyond character-counting.

- **Prototype 3 (Final SentinelVault):** The integration of the Web Crypto API for SHA-1 hashing and the HIBP API. This prototype handles the "k-Anonymity" logic: sending only the first 5 characters of a hash to a server and verifying the results locally.

5. Expected Outcomes

It is expected that rule-based systems will incorrectly label "complex" but common passwords (e.g., Qwerty!123) as strong. The SentinelVault system is expected to correctly identify these as "Weak" or "Compromised" due to its ability to recognize patterns and detect them in previous data breaches.

6. Resources and Materials

- **Hardware:** Personal laptop.
- **Software/Languages:** HTML5, CSS3, JavaScript (ES6).
- **Development Tools:** Visual Studio Code, GitHub Desktop.
- **Libraries/APIs:** zxcvbn.js, Have I Been Pwned API.

7. Ethics and Plagiarism

- **Ethics:** No real-world user data or personally identifiable information (PII) is collected or stored. The project uses a "Privacy-First" architecture where all sensitive matching happens in the user's local browser.
- **Plagiarism:** All external libraries (zxcvbn) and APIs (HIBP) are acknowledged and cited using the Harvard referencing style as per the Eskom Expo Guidebook.

8. Timeline (Milestones)

- **Feb/March 2026:** Initial research and Research Plan approval.
- **April 2026:** Development of Prototype 1 (UI) and Prototype 2 (Local Heuristics).
- **April 2026:** Integration of API/Hashing logic (Prototype 3) and final debugging.
- **May 2026:** Final testing, results analysis, and completion of the Project Report/Poster.